

Quantum sized
Quantum physics studies phenomena that occur at extreme measurements. Subatomic particles cannot be observed directly but they can be studied through experiments that observe their effects.



PLANCK LENGTH
 10^{-35}m

This is the smallest unit of length possible in current physics theories. At lengths at or below the Planck length, current theories of physics break down and can no longer make sensible predictions.

VANISHINGLY SMALL

While the largest atoms have a diameter of about half a nanometer (billionth of a meter)—less than $1/100,000$ th the width of a human hair—most of their volume is a sparse cloud filled with electrons around a dense central nucleus. Diameters of atomic nuclei are typically a few femtometers (million billionths of a meter), and it is usually at around these scales (and even smaller ones) that strange quantum behaviors become apparent. The smallest distance that makes physical sense is a Planck unit of length (see pp.140–41).